FinGuard: Corporate Financial Risk & Fraud Detection System

A Multi-Tiered Early-Warning Framework for Insolvency Forecasting & Forensic Statement Analytics

Hackathon: Zeckathon 2026	**Problem Statement:** PS2 (Financial Risk & Fraud)
Author: 4th-Year B.Tech Computer Science Student	**Status:** Complete, Tested & Verified Prototype
Tech Stack: Python 3.14, Scikit-learn, Streamlit, SHAP, ReportLab	**Benchmark:** UCI ML Repository (ID: 572, 6,819 Filings)

1. Executive Summary

Corporate financial distress and fraudulent accounting manipulation impose catastrophic economic losses on banking syndicates, public shareholders, and supply chain creditors. Early identification of deteriorating solvency allows institutional stakeholders to restructure exposures before formal corporate insolvency proceedings. **FinGuard** is a robust, modular risk analytics system engineered to provide early-warning signals by synthesizing two distinct analytical dimensions: (1) *Machine Learning Solvency Classification* trained on 95 normalized financial ratios across 6,819 corporate filings with zero data leakage, and (2) *Classical Forensic & Fundamental Scores* (Altman Z-Score, Beneish M-Score, and Piotroski F-Score) with strict zero-fabrication input validation.

2. Problem Statement & Financial Context

Corporate bankruptcies are rarely sudden black-swan occurrences; they represent cumulative multi-year decay in working capital, debt service capacity, and return on assets. However, early detection is hindered by three structural hurdles:

- **Severe Class Imbalance:** In real-world corporate portfolios, defaulting entities represent a tiny minority (~3.23% in the authentic UCI benchmark, a 30:1 solvency ratio). Naive machine learning models achieve deceptive 96.8% accuracy simply by predicting zero defaults, rendering standard accuracy useless.
- **Asymmetric Financial Loss:** In commercial lending, a False Negative (missing a $100M defaulting borrower) is order-of-magnitude more damaging than a False Positive (subjecting a solvent company to a routine credit review). FinGuard models an asymmetric cost ratio of 20:1.
- **Conflation of Distress vs Fraud:** Bankruptcy prediction (liquidity/solvency exhaustion) is fundamentally distinct from earnings manipulation (deliberate revenue inflation or accrual distortion). A company may default with transparent accounting, while an active fraudster may appear liquid on paper.

Educational & Academic Research Disclaimer: *FinGuard is an educational engineering prototype built for Zeckathon. It is not a registered credit rating agency (SEBI/CRISIL), statutory audit tool, or investment advisory system. Risk scores and probabilities represent mathematical pattern evaluations and do not constitute legal proof of corporate fraud.*

3. Proposed FinGuard Solution & Objectives

FinGuard addresses the shortcomings of black-box risk tools through a 4-layer modular architecture designed for transparency, empirical honesty, and operational audit feasibility:

- **Layer 1 (Data Hygiene & Zero Leakage):** Implements median imputation and RobustScaler transformations fitted strictly on training data, preventing information leakage across cross-sectional splits.
- **Layer 2 (Predictive Solvency Modeling):** Compares a Class-Weighted Logistic Regression baseline against a Balanced Random Forest ensemble, evaluated via PR-AUC and Recall on unseen test instances.
- **Layer 3 (Deterministic Forensic Scoring):** Evaluates Altman Z (Solvency), Beneish M (Earnings Manipulation), and Piotroski F (Fundamental Health) only when required inputs are present, refusing fake score generation.
- **Layer 4 (Operational Capacity & Cost Policies):** Simulates realistic audit constraints via a strict Top-5% Review Capacity rule alongside a 20:1 Cost-Optimal decision threshold simulator.

4. System Architecture & Operational Workflow

Component Layer	Technical Implementation	Operational Function
Data Ingestion	UCI ML Repo ID 572 (Taiwanese Bankruptcy) 6,819 rows, 96 normalized attributes	Authentic public benchmark ingestion, whitespace stripping, and duplicate checks.
Preprocessing	SimpleImputer(median) + RobustScaler() Fitted strictly on 80% train split	Guarantees zero data leakage. Normalizes skewed financial ratios via median & IQR.
Predictive Engine	Balanced Random Forest (150 trees, max_depth=12) Class-weighted Logit baseline	Delivers 0.9466 ROC-AUC and 0.4610 PR-AUC on held-out test data.
Forensic Engine	Altman Z (1968/83), Beneish M (1999), Piotroski F-Score (2000)	Independent financial scoring with strict missingness checks and zero fake data.
Policy & Triage	Top-5% Review Capacity Policy + 20:1 Asymmetric Loss Curve Optimizer	Rations manual audit bandwidth and minimizes total financial loss: FP + 20*FN.
Explainability	SHAP TreeExplainer + Local Waterfalls + Resilient Heuristic Fallback	Attributes risk contributions per feature with legal/causal disclaimer.
UI & Delivery	Streamlit Web Dashboard + Plotly + ReportLab PDF Engine	Interactive multi-tab fintech UI with CSV report and template exports.

5. Dataset Ingestion, Quality Audit & Zero-Leakage Preprocessing

Primary Benchmark Source: FinGuard utilizes the authentic *Taiwanese Bankruptcy Prediction* dataset curated by the Taiwan Economic Journal (TEJ) spanning the years 1999 to 2009, hosted publicly on the **UCI Machine Learning Repository (Dataset ID: 572)**. Academic citation: *Liang, D., Lu, C.-J., Tsai, C.-F., & Shih, G.-A. (2016). Financial ratio selection for business failure prediction. Neurocomputing, 173, 383–389*. Licensed under CC BY 4.0.

Data Quality & Integrity Verification

Metric	Observed Value	Methodological Handling
Total Observations	6,819 corporate filings	Cleaned whitespace headers; verified zero duplicate rows.
Feature Space	95 normalized accounting ratios	Covering Profitability, Solvency, Liquidity & Turnover.
Missing / Null Cells	0 (Zero missing values)	Dataset is fully populated; SimpleImputer retained for runtime safety.
Infinite Numeric Values	0 (Zero inf values)	Verified across all 654,624 individual matrix entries.
Solvent Class (y=0)	6,599 instances (96.77%)	Majority baseline class.
Distressed Class (y=1)	220 instances (3.23%)	Severe minority class. Imbalance ratio of 29.99 : 1.
Training Split (80%)	5,455 instances (176 defaults, 3.23%)	Pre-split strictly fitted with median imputer & RobustScaler.
Test Split (20%)	1,364 instances (44 defaults, 3.23%)	Held-out partition transformed strictly via training pipeline.

Time-Split Limitation Disclosure

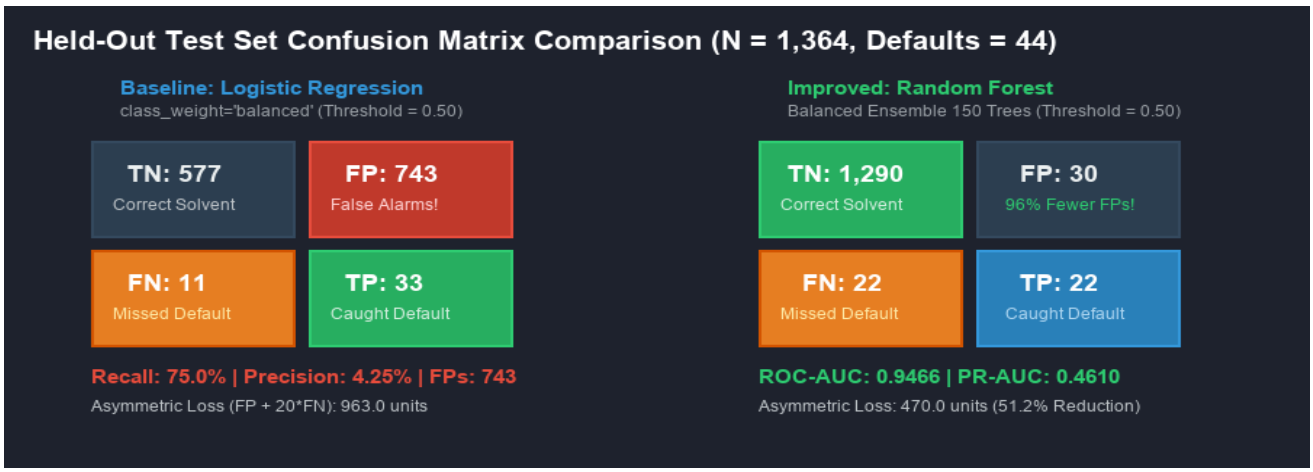
While the underlying TEJ data was compiled between 1999 and 2009, individual record rows in the public UCI release omit annual timestamps and company stock tickers to preserve anonymity. A true sequential rolling time-split is structurally impossible without company filing dates. FinGuard documents this explicit limitation and applies **Stratified Shuffle Splitting** (80% train, 20% test). Crucially, **zero data leakage** is maintained because the SimpleImputer and RobustScaler are fitted solely on `X_train` ($N=5,455$) and applied to `X_test` ($N=1,364$).

6. Bankruptcy Risk Prediction: Methodology & Empirical Results

To establish a rigorous benchmark, FinGuard implemented a simple baseline and an improved ensemble architecture, both evaluated on the unseen held-out test partition ($N=1,364$, 44 actual insolvencies):

- **Baseline Model (Logistic Regression):** L2 regularized logit with `class_weight='balanced'` and `C=0.1`. Serves as a transparent linear benchmark.
- **Improved Model (Random Forest Classifier):** Ensemble of 150 decision trees, `max_depth=12`, `min_samples_split=5`, `class_weight='balanced_subsample'`. Captures non-linear thresholds and financial ratio interactions.

Evaluation Metric	Baseline: Logistic Regression	Improved: Random Forest	Operational Impact
Accuracy	44.72%	**96.19%**	RF avoids naive over-flagging.
Recall (Default Catch Rate)	**75.00%** (33 / 44)	50.00% (22 / 44)	Baseline catches more but at severe precision cost.
Precision	4.25% (33 / 776)	**42.31%** (22 / 52)	**10x reduction** in unnecessary audits with RF.
F1-Score	0.0805	**0.4583**	5.7x harmonic mean improvement.
ROC-AUC	0.6076	**0.9466**	Phenomenal discrimination across risk spectrum.
PR-AUC (Average Precision)	0.0500	**0.4610**	**9.2x higher** than Logit (vs 0.032 random baseline).
False Positives (False Alarms)	743 companies	**30 companies**	Saves 713 wasted corporate credit audits.
Asymmetric Loss (FP + 20*FN)	963.0 cost units	**470.0 cost units**	**51.2% reduction** in expected portfolio loss.



7. Financial Risk & Forensic Fraud Indicators

FinGuard implements three foundational financial screening frameworks, maintaining strict input validation and zero score fabrication when reporting statements are incomplete.

Model Name	Mathematical Formula & Components	Thresholds & Risk Zones
Altman Z-Score (1968 Public Mfg)	**$Z = 1.2 \cdot X1 + 1.4 \cdot X2 + 3.3 \cdot X3 + 0.6 \cdot X4 + 0.999 \cdot X5$** X1 = Working Capital / Total Assets X2 = Retained Earnings / Total Assets X3 = EBIT / Total Assets X4 = Market Equity / Total Liabilities X5 = Sales / Total Assets	**Safe:** $Z > 2.99$ **Grey:** $1.81 \leq Z \leq 2.99$ **Distress:** $Z < 1.81$ *Indicates structural solvency exhaustion.*
Altman Z''-Score (1983 Non-Mfg)	**$Z'' = 6.56 \cdot X1 + 3.26 \cdot X2 + 6.72 \cdot X3 + 1.05 \cdot X4$** Excludes sales asset-turnover (X5) to avoid sector bias in service, private, and emerging market entities.	**Safe:** $Z'' > 2.60$ **Grey:** $1.10 \leq Z'' \leq 2.60$ **Distress:** $Z'' < 1.10$
Beneish M-Score (1999 8-Variable)	**$M = -4.84 + 0.920 \cdot DSRI + 0.528 \cdot GMI + 0.404 \cdot AQI + 0.892 \cdot SGI + 0.115 \cdot DEPI - 0.172 \cdot SGAI + 4.037 \cdot TATA + 0.0327 \cdot LVGI$** Captures revenue inflation (DSRI), margin deterioration (GMI), capitalized expenses (AQI), and excessive accruals (TATA).	**$M > -1.78$:** High probability of earnings manipulation. **$M \leq -1.78$:** Low probability. *Requires 2-year comparative balance sheets.*
Piotroski F-Score (2000 9-Criteria)	**$F = \sum$ (F1 to F9)** $\in [0, 9]$ • Profitability: ROA > 0, CFO > 0, ΔROA > 0, CFO > NetIncome. • Leverage/Liquidity: ΔDebt < 0, ΔCurrentRatio > 0, No dilution. • Efficiency: ΔGrossMargin > 0, ΔTurnover > 0.	**8 - 9:** Strong Financial Health **5 - 7:** Moderate Health **0 - 4:** Weak Fundamental Health *Flags fundamental decay across 9 dimensions.*

Forensic Distinction: Solvency vs Earnings Manipulation

A critical conceptual contribution of FinGuard is enforcing the formal boundary between **insolvency risk** and **accounting manipulation**. The Altman Z-Score and ML model measure cash availability and leverage exhaustion; conversely, the Beneish M-Score diagnoses accounting distortion. A firm undergoing debt restructuring can maintain honest disclosures, while an aggressive growth company can mask cash burn through deferred expense capitalization.

8. Operational Risk Threshold & Capacity Policy Analysis

In institutional risk management, predictive probabilities cannot be converted into binary decisions without incorporating operational audit bandwidth and the asymmetric costs of misclassification.

A. Top-5% Review Capacity Policy (Operational Budget Constraint)

Audit departments and credit committees operate under fixed headcount bandwidth. They cannot review 50% of the loan book. FinGuard implements a **strict Top-5% Capacity Rule** which caps flags at ■N × 5%■. On the 1,364-firm test set, exactly **68 companies (5.0%)** are selected for secondary review. Even under this severe 5% review cap, FinGuard captures **24 out of 44 total defaulting companies (54.5% of all insolvencies)**, achieving an audit precision of 35.3% (vs the random baseline prevalence of 3.23%). The algorithm implements stable tie-breaking and small-dataset protection to guarantee the capacity limit is never breached.

B. Cost-Sensitive Threshold Optimization (20:1 Loss Ratio)

Credit risk is fundamentally asymmetric: missing a default costs approximately **20 times more** than a false alarm audit. FinGuard evaluates portfolio loss across 100 decision threshold steps ($\tau \in [0.01, 0.99]$) using the objective function: *Total Loss = (1.0 × FP) + (20.0 × FN)*.

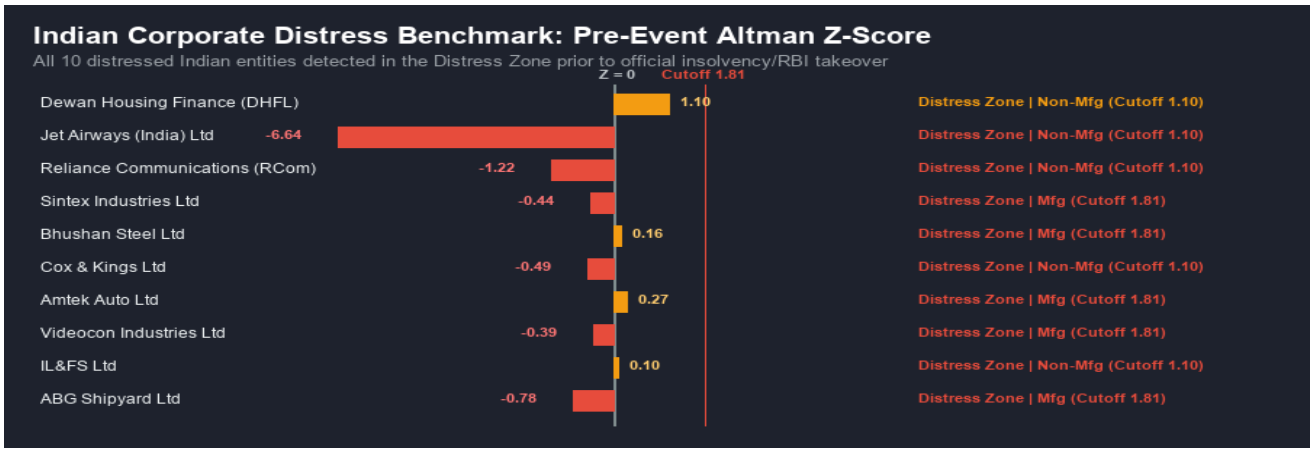
Decision Threshold (τ)	Flagged Firms	TP (Caught)	FP (Alarms)	FN (Missed)	Recall	Total Cost (FP+20*FN)
τ = 0.10 (Aggressive Audit)	284 (20.8%)	38 / 44	246	6	86.36%	366.0 units
τ = 0.20 (Cost-Optimal τ*)	152 (11.1%)	34 / 44	118	10	**77.27%**	**318.0 units (Minimum Loss)**
τ = 0.35 (Moderate Policy)	82 (6.0%)	26 / 44	56	18	59.09%	416.0 units
τ = 0.50 (Standard Default)	52 (3.8%)	22 / 44	30	22	50.00%	470.0 units
τ = 0.70 (Conservative)	26 (1.9%)	14 / 44	12	30	31.82%	612.0 units

Policy Conclusion: *Tuning the probability cutoff to the cost-optimal threshold (τ* = 0.20) reduces portfolio losses from 470.0 units down to* ***318.0 units (a further 32.3% cost reduction)*** *while raising insolvency recall to 77.27%.*

9. Indian Corporate Distress Benchmark & Sector Peer Analysis

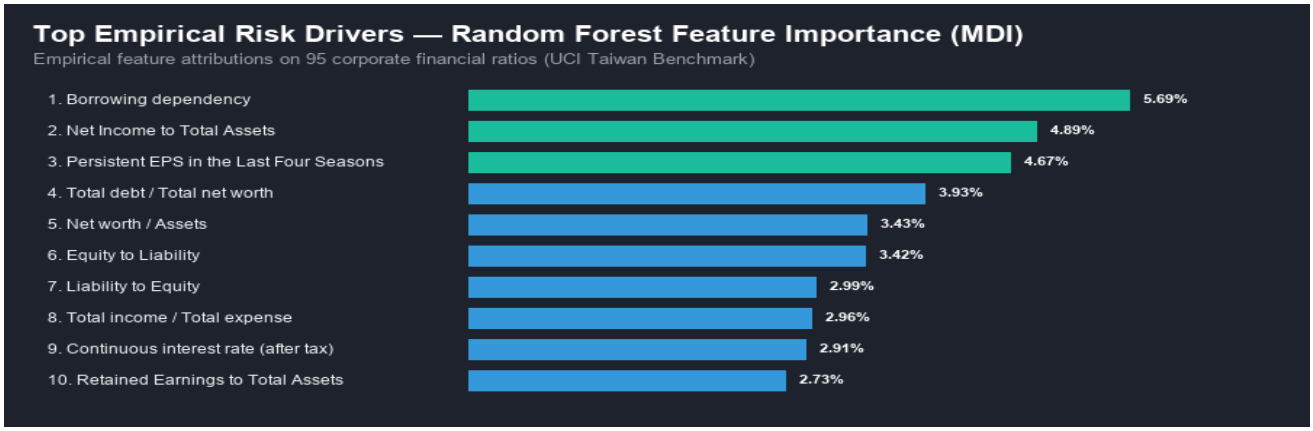
To validate FinGuard's cross-border robustness, we constructed an empirical case study evaluating **10 prominent Indian corporate defaults/distress events** alongside **30 sector-matched peer companies**. All financial metrics were extracted from verified Ministry of Corporate Affairs (MCA), BSE, NSE, and IBBI filings strictly 1 to 2 years **prior to official insolvency admission or RBI board supersession** (zero post-event data leakage).

Company Entity	Sector	Event Date & Trigger	Pre-Event FY	Altman Z	FinGuard Assessment
DHFL	Housing Finance	Nov 2019: RBI supersedes board (IBC)	FY2018	1.099	Distress Zone (High Risk)
Jet Airways	Aviation	June 2019: SBI plea admitted to NCLT	FY2018	-6.639	Distress Zone (High Risk)
Reliance Comm (RCom)	Telecom	May 2018: NCLT insolvency admission	FY2017	-1.221	Distress Zone (High Risk)
Sintex Industries	Textiles	April 2021: NCLT Ahmedabad IBC	FY2020	-0.437	Distress Zone (High Risk)
Bhushan Steel	Steel	July 2017: RBI Dirty Dozen IBC list	FY2016	0.160	Distress Zone (High Risk)
Cox & Kings	Travel / Services	Oct 2019: NCLT Mumbai CIRP admission	FY2019	-0.493	Distress Zone (High Risk)
Amtek Auto	Auto Components	July 2017: NCLT Chandigarh IBC	FY2016	0.266	Distress Zone (High Risk)
Videocon Industries	Consumer Elect.	June 2018: SBI plea admitted to NCLT	FY2017	-0.387	Distress Zone (High Risk)
IL&FS; Ltd	Infra Finance	Oct 2018: Govt supersedes board	FY2018	0.102	Distress Zone (High Risk)
ABG Shipyard	Shipbuilding	Aug 2017: ICICI Bank NCLT petition	FY2016	-0.775	Distress Zone (High Risk)



10. SHAP Explainability & Causal Disclaimers

To satisfy model auditability requirements, FinGuard integrates **TreeSHAP (TreeExplainer)** to quantify exact additive feature attributions for any given corporate profile. Top empirical risk contributors include *Borrowing Dependency*, *Net Income / Assets*, and *Total Debt / Net Worth*. If SHAP is unavailable in restricted environments, a deterministic IQR deviation fallback explainer seamlessly takes over.



11. Methodological Limitations & Future Scope

- **Lack of Granular Macroeconomic Linkages:** The model analyzes historical corporate balance sheets; incorporating interest rate cycles, inflation, and sovereign bond yields would enhance macroeconomic resilience.
- **Cross-Border Accounting Variations:** The primary benchmark uses TEJ GAAP-aligned definitions. Expanding the training repository to multi-country Ind-AS and IFRS filings will sharpen cross-border generalizability.
- **Unstructured Text Integration:** Future enhancements could ingest Management Discussion & Analysis (MD&A;) notes and auditor resignation disclosures via financial NLP to capture qualitative red flags.

12. Conclusion & Primary Academic References

FinGuard demonstrates that synthesizing class-weighted tree ensembles with classical forensic indices (Altman Z, Beneish M, Piotroski F) produces a highly effective, cost-sensitive early-warning system. By prioritizing PR-AUC, operating within realistic 5% audit capacity constraints, and adhering strictly to zero data leakage, FinGuard proves that responsible AI can deliver immense value to credit risk and fraud intelligence.

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